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AC31 – Despliegue de aplicaciones web

Tarea 1 – Fork y clonación del repositorio

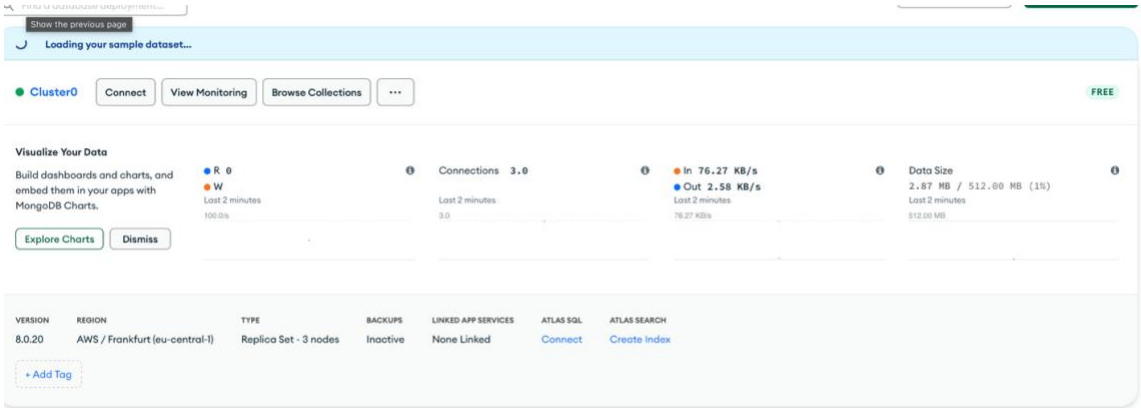
1. Se hace un fork del repositorio en GitHub y se clona el fork en la máquina local con git clone.

```
marc@Gumis-MacBook-Pro ac31 % git clone https://github.com/Rayo25Gumi/M-AC1API.git
cd M-AC1API
```

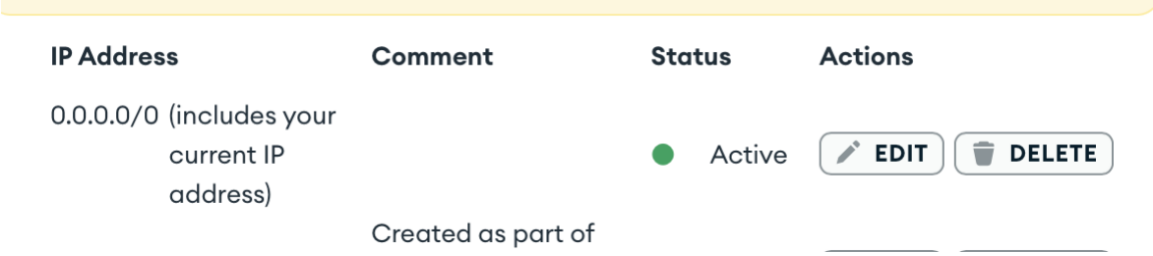
```
[marc@Gumis-MacBook-Pro M-AC1API %  ls
 app.js          config          db              package.json    routes
 auth            controllers     models          README.md       views
 marc@Gumis-MacBook-Pro M-AC1API %
```

Tarea 2 – Configuración de MongoDB Atlas

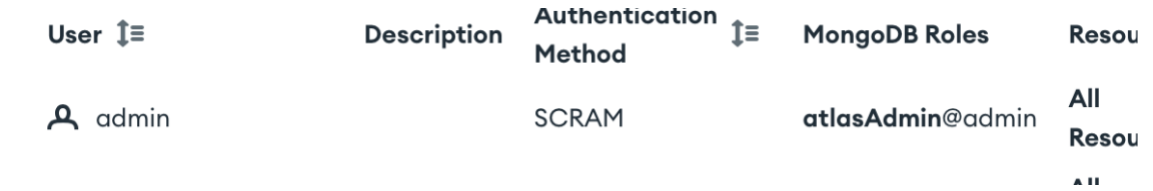
1. Se crea un clúster gratuito en MongoDB Atlas y una base de datos llamada products.



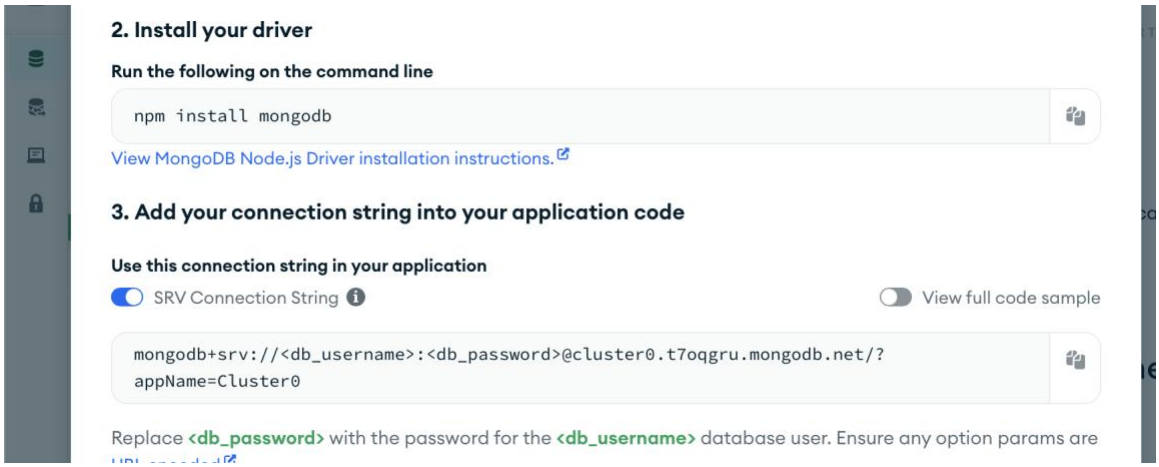
2. Se crea un nuevo usuario con permisos de administrador.



3. Se añade la IP local a la whitelist del clúster.



4. Se obtiene la URI de conexión y se añade a config/config.js.



Tarea 3 – Instalación de dependencias y prueba en local

1. Se instalan las dependencias con npm install.

2. Se ejecuta la aplicación con nodemon app.js y se comprueba que funciona en http://localhost:3000/api.

```
.gitignore
node_modules
views/styles/styles.css
```

3. Se registra un nuevo usuario vía POST /users/register y se obtiene el apikey.

```
package.json > ...
{
  "name": "productsapi",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  ▷Debug
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^4.18.2",
    "mongodb": "^4.14.0",
    "pug": "^3.0.2"
  }
/Año 2/Desplegament/3r/ac31/M-AC1API/app.js
```

4. Se crean productos de prueba vía POST /api/products.

```
marc@Gumis-MacBook-Pro M-AC1API %
marc@Gumis-MacBook-Pro M-AC1API % npm install

added 207 packages, and audited 208 packages in 9s

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
marc@Gumis-MacBook-Pro M-AC1API %
```

5. Se consulta la lista de productos vía GET /api/products.

```
  found 0 vulnerabilities
marc@Gumis-MacBook-Pro M-AC1API % npm install -g nodemon


added 28 packages in 3s

5 packages are looking for funding
  run `npm fund` for details
marc@Gumis-MacBook-Pro M-AC1API %
```

6. Se observa que el proyecto utiliza el motor de plantillas Pug, configurado en app.js.

```
Error: listen EADDRINUSE: address already in use :::5000
    at Server.setupListenHandle [as _listen2] (node:net:2016:16)
    at listenInCluster (node:net:2073:12)
    at Server.listen (node:net:2178:7)
    at app.listen (/Users/marc/Documents/Clase/Año 2/Desplegament/3r/ac31/M-AC1API/node_modules/express/lib/application.js:635:24)
    at Object.<anonymous> (/Users/marc/Documents/Clase/Año 2/Desplegament/3r/ac31/M-AC1API/app.js:17:5)
    at Module._compile (node:internal/modules/cjs/loader:1811:14)
    at Module._extensions..js (node:internal/modules/cjs/loader:1942:10)
    at Module.load (node:internal/modules/cjs/loader:1532:32)
    at Module._load (node:internal/modules/cjs/loader:1334:12)
    at wrapModuleLoad (node:internal/modules/cjs/loader:255:19)
Emitted 'error' event on Server instance at:
    at emitErrorNT (node:net:2052:8)
    at process.processTicksAndRejections (node:internal/process/task_queues:90:21) {
  code: 'EADDRINUSE',
  errno: -48,
  syscall: 'listen',
  address: '::',
  port: 5000
}

Node.js v25.6.1
[nodemon] app crashed - waiting for file changes before starting...
```

7. Se suben los cambios al repositorio (sólo config.js ha cambiado).

```
Node.js v25.6.1
[nodemon] app crashed - waiting for file changes before starting...

[nodemon] restarting due to changes...
[nodemon] starting `node app.js`
server is listening on port 5000
[nodemon] restarting due to changes...
[nodemon] starting `node app.js`
server is listening on port 3000
```

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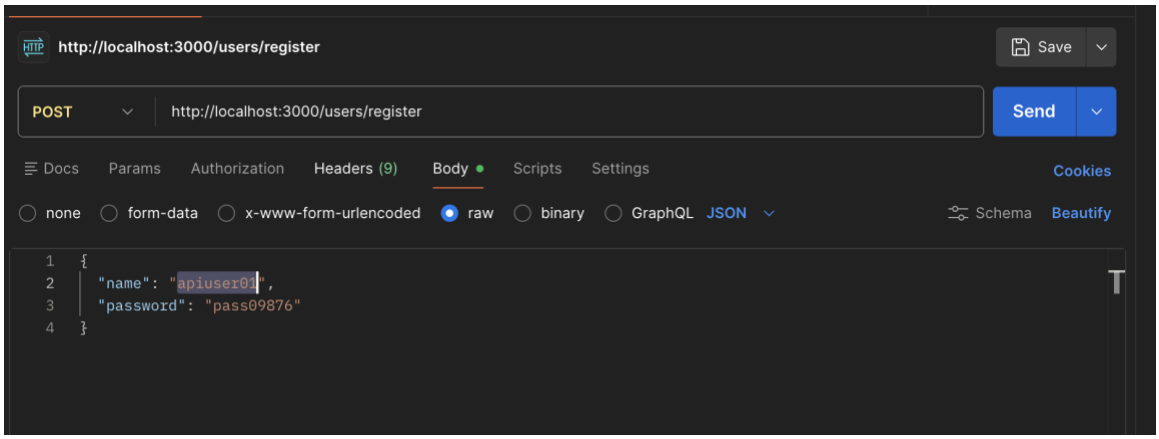
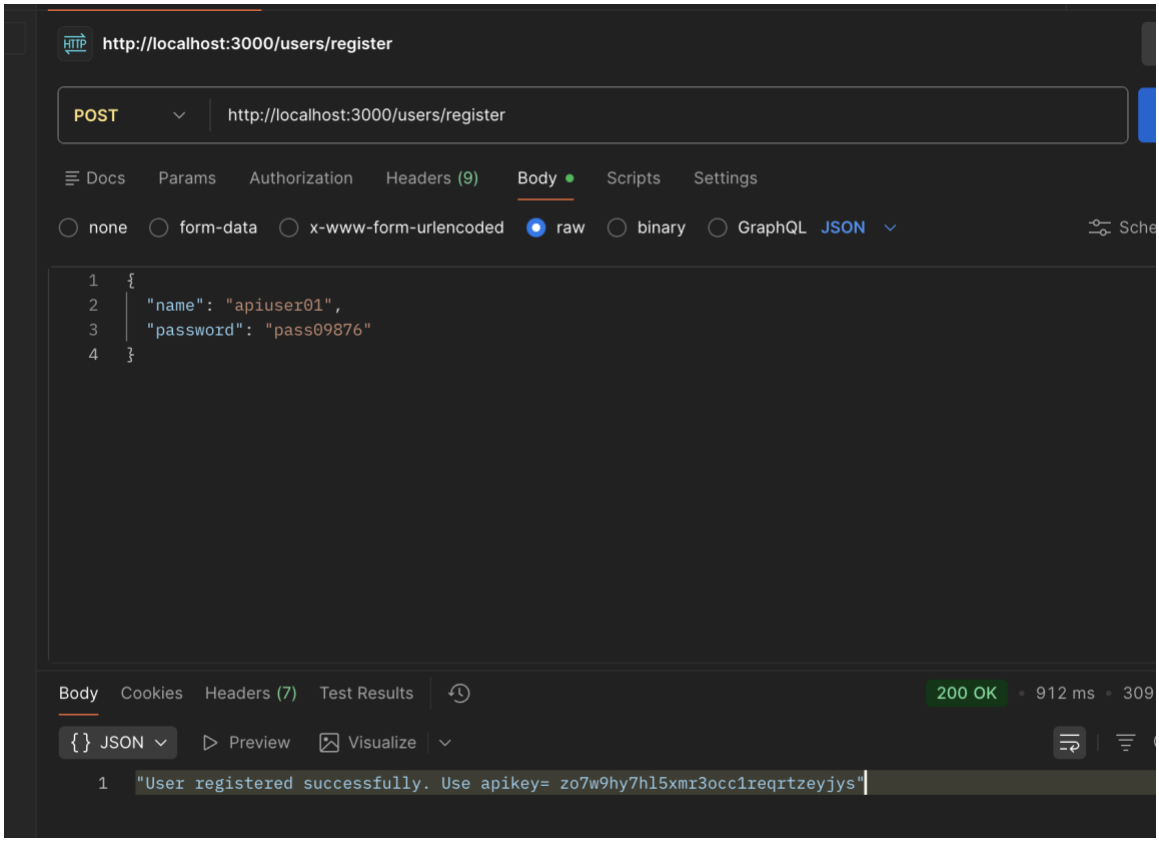
Tarea 4 – Desplegament online a Render

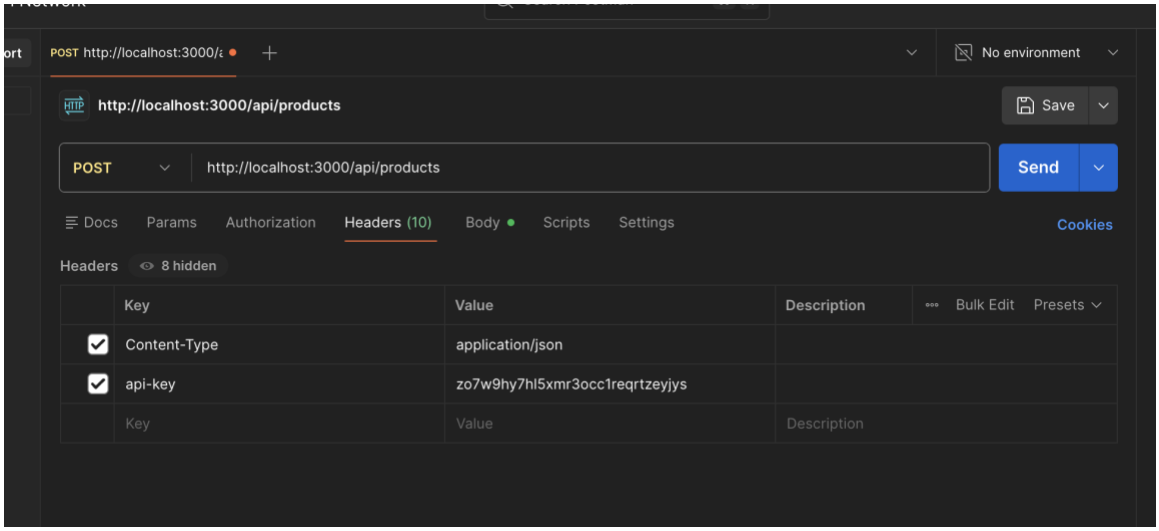
1. Se crea una cuenta en render.com y se conecta con GitHub.



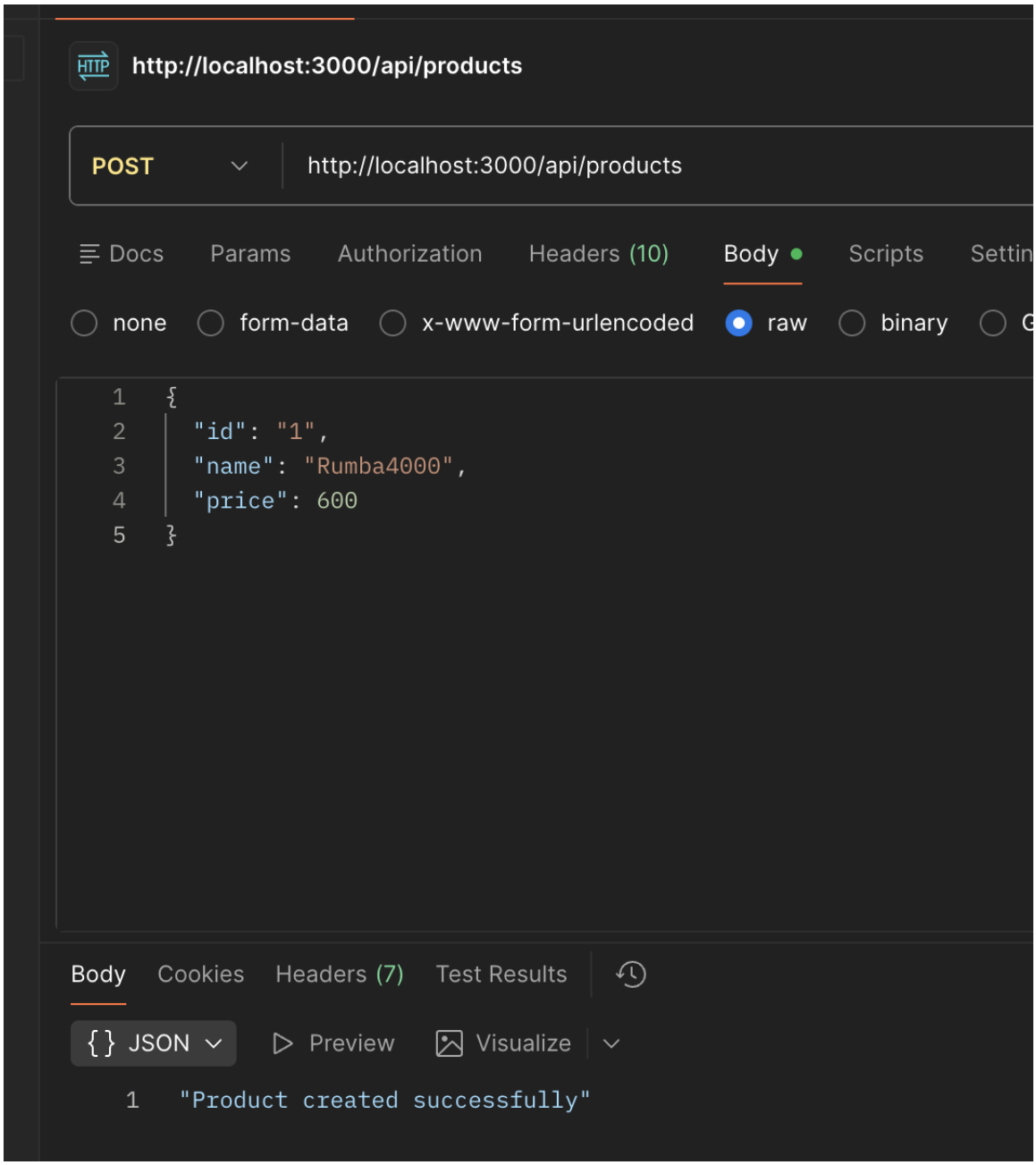
2. Se crea un nuevo Web Service apuntando al repositorio fork.



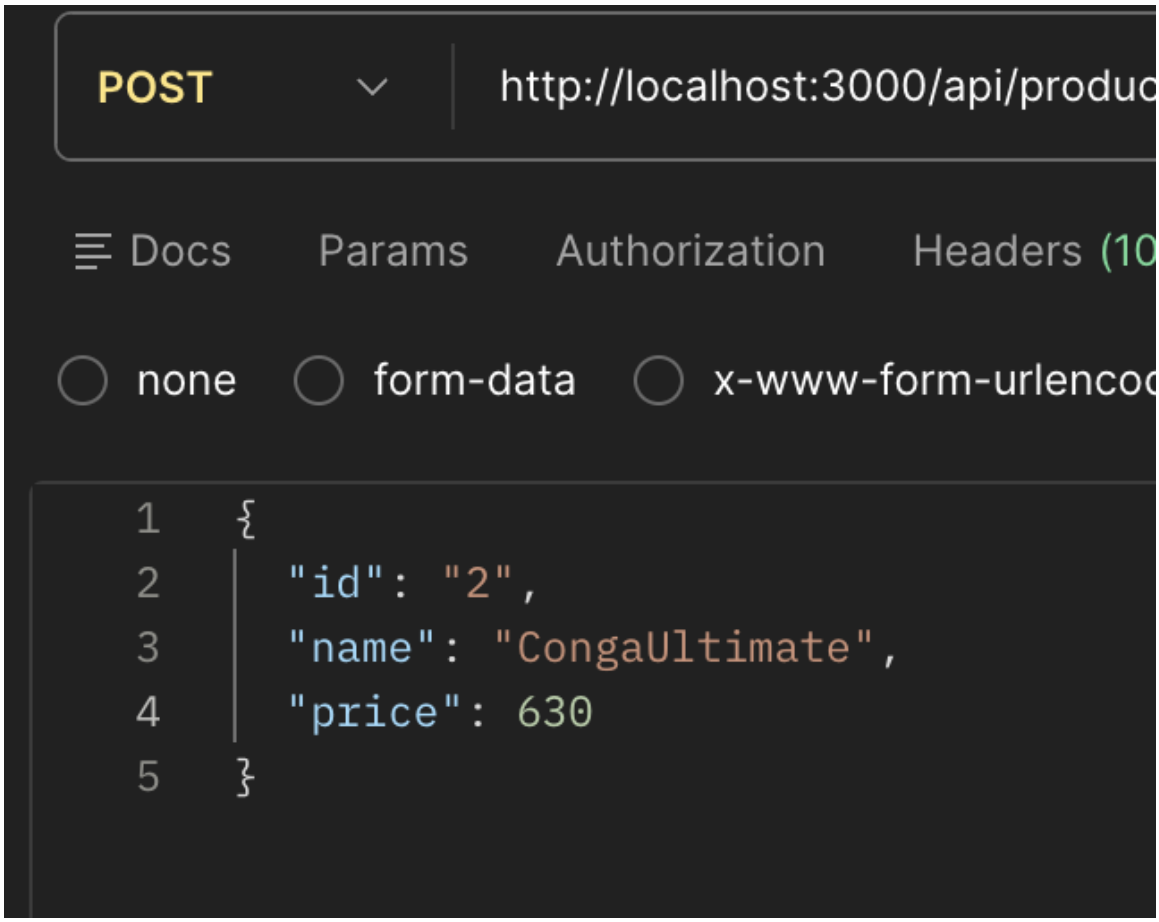
3. Comando de compilación: instalación de npm | Comando inicio: node --unhandled-rejections=app.js estricta



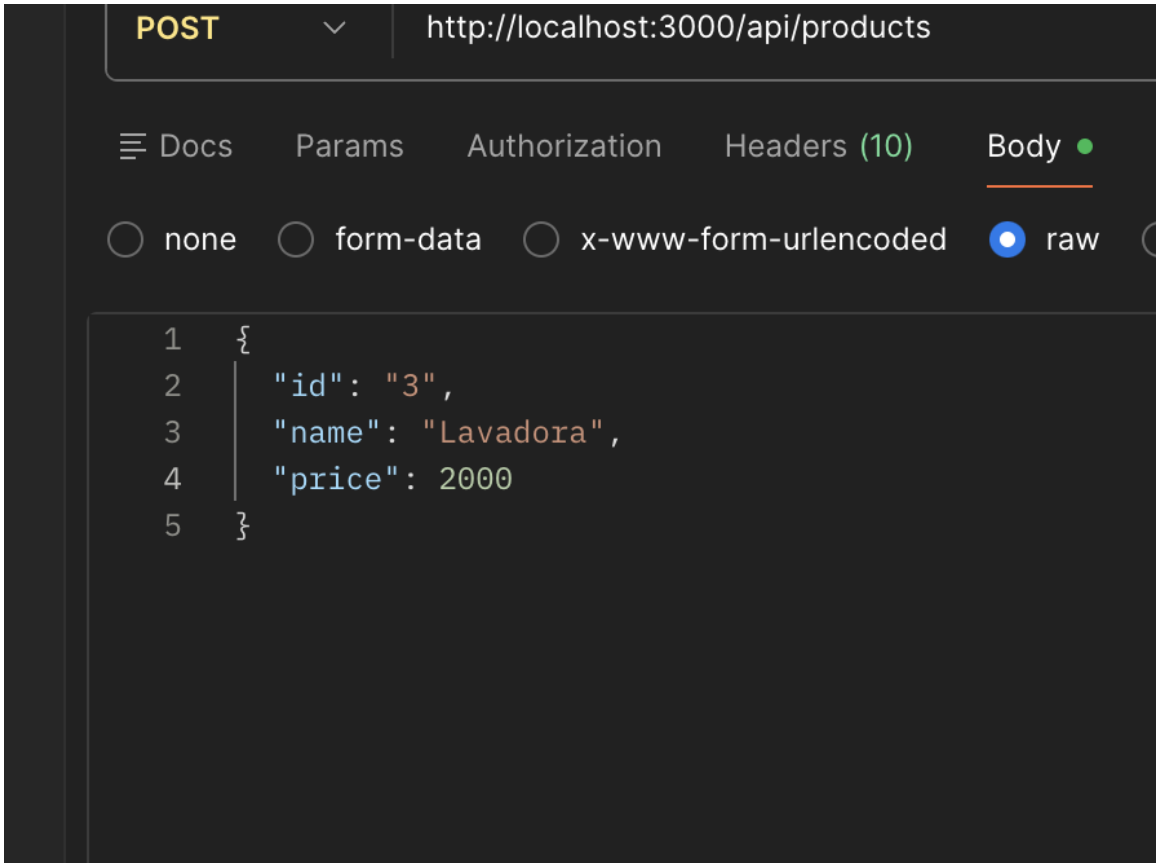
4. Se captura el log del proceso de deployment.



5. Se añaden las IPs de Render a la whitelist de Atlas para que los endpoints funcionen.



6. Se comprueba el funcionamiento completo de la API desde la URL de Render.

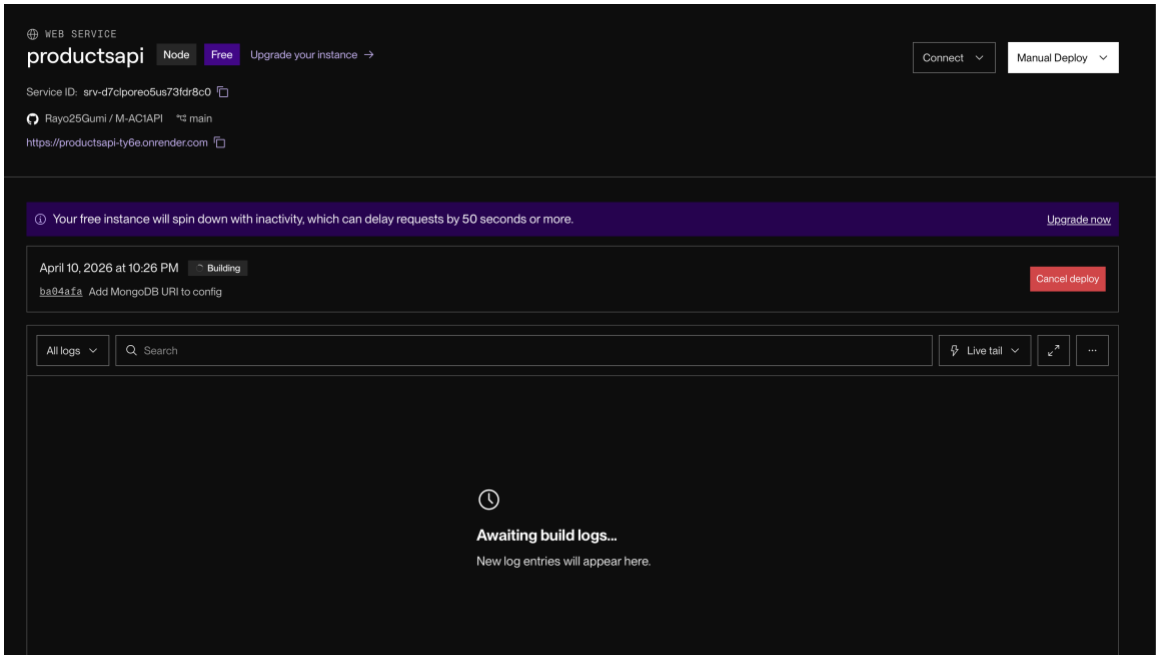


Tarea 5 – Autodeploy con slugify

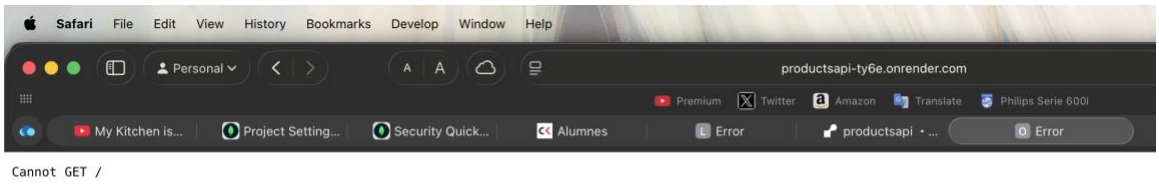
1. S'instal·la slugify: npm install slugify.

```
Product created successfully
^C%
marc@Gumis-MacBook-Pro M-AC1API % git add config/config.js
git commit -m "Add MongoDB URI to config"
git push origin main
[main ba04afa] Add MongoDB URI to config
 1 file changed, 3 insertions(+), 2 deletions(-)
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 428 bytes | 428.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Rayo25Gumi/M-AC1API.git
   418b820..ba04afa  main -> main
marc@Gumis-MacBook-Pro M-AC1API %
```

2. Se usa slugify para slugizar con los mensajes de bienvenida del servidor Express a app.js.



3. Se suben los cambios al repositorio y Render lanza automáticamente un nuevo deploy.



4. Se comprueba en los logs de Render que el mensaje aparece slugizado correctamente.

https://productsapi-ty6e.onrender.com/api/products

POST https://productsapi-ty6e.onrender.com/api/products

Docs Params Authorization Headers (10) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "id": "4",
3   "name": "test",
4   "price": 200
5 }
```

Body Cookies Headers (14) Test Results 20

JSON Preview Visualize

```
1 "Product created successfully"
```

Tarea 6 – Automatización de la compilación SASS

1. Se instala sass como dependencia de desarrollo (node-sass está obsoleto): npm install --save-dev sass.

```
418b820..ba04afa  main -> main
marc@Gumis-MacBook-Pro M-AC1API % npm install slugify

added 1 package, and audited 209 packages in 915ms

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
marc@Gumis-MacBook-Pro M-AC1API %
```

0 Connect Live Share

2. Es compila el SCSS a CSS: npx sass ./views/styles/main.scss ./views/styles/styles.css

```
app.js > ...
const express = require('express')
const slugify = require('slugify')

const products_routes = require('./routes/products.js')

//Server instantiation
const app = express()

//Server configuration: template engine
app.set('views', './views');
app.set('view engine', 'pug');
app.use(express.static('/views'));

//Midleware
app.use(express.json())
app.use('/', products_routes)

//Server startup
app.listen(3000, () => {
    console.log(slugify('server is listening on port 3000', { replacement: '*' }))
})
```

3. La página de bienvenida ya aparece con estilos aplicados.

```
  </body>
  </html>
● marc@Gumis-MacBook-Pro M-AC1API % git add .
  git commit -m "Add slugify to welcome message"
  git push origin main
  [main 83f003c] Add slugify to welcome message
   3 files changed, 2856 insertions(+), 3 deletions(-)
   create mode 100644 package-lock.json
  Enumerating objects: 8, done.
  Counting objects: 100% (8/8), done.
  Delta compression using up to 8 threads
  Compressing objects: 100% (5/5), done.
  Writing objects: 100% (5/5), 22.82 KiB | 22.82 MiB/s, done.
  Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
  remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
  To https://github.com/Rayo25Gumi/M-AC1API.git
     ba04afa..83f003c  main -> main
  marc@Gumis-MacBook-Pro M-AC1API %
Connect   Live Share
```

4. Se añaden los scripts al package.json: build:css y build.

Filter events 31

Deploy live for ba04afa: Add MongoDB URI to config

5. Se hace algún cambio en el SASS y se recompila con npm run build:css.

```
npm error Build failed with error code: 1
npm error A complete log of this run can be found in: /Users/marc/.npm/_logs/2026-04-10T20_45_49_233Z-debug-0.log
marc@Gumis-MacBook-Pro M-AC1API % npm install --save-dev sass

added 12 packages, and audited 221 packages in 3s

32 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
marc@Gumis-MacBook-Pro M-AC1API % npx sass ./views/styles/main.scss ./views/styles/styles.css
marc@Gumis-MacBook-Pro M-AC1API %
Connect  Live Share                                  Watch Sass  Ln 1, Col 1  Spa
```

6. S'actualiza el comando de build: Render a npm run build.

```
package.json > ...
1  {
2    "name": "productsapi",
3    "version": "1.0.0",
4    "description": "",
5    "main": "index.js",
       Debug
6    "scripts": {
7      "test": "echo \"Error: no test specified\" && exit 1",
8      "build:css": "sass ./views/styles/main.scss ./views/styles/styles.css",
9      "build": "npm install && npm run build:css"
10   },
11   "author": "",
12   "license": "ISC",
13   "dependencies": {
14     "express": "^4.18.2",
15     "mongodb": "^4.14.0",
16     "pug": "^3.0.2",
17     "slugify": "^1.6.9"
18   },
19   "devDependencies": {
20     "sass": "^1.99.0"
21   }
22 }
```

7. Se suben los cambios y Render genera el CSS automáticamente durante el deploy.

```
JS config.js    JS products.js    JS app.js    main.scss M ×
views > styles > main.scss > body
1   body{
2       display: flex;
3       flex-direction: column;
4       align-items: center;
5   }
6   h2{
7       color: rgb(51, 255, 0);
8   }
9   table{
10      border-collapse: collapse;
11  }
12  td{
13      border: solid 2px blue;
14      padding: 10px;
15  }
16  th{
17      border: solid 2px blue;
18      padding: 10px;
19  }
```

Tarea 7 – Ocultación de la URI (variables de entorno)

1. Se define la variable de entorno MONGOURI en la terminal.

```
83f003c..593afae  main -> main
marc@Gumis-MacBook-Pro M-AC1API % export MONGOURI='mongodb+srv://admin:GVSMLZJXW0VFWU8j@cluster0.t7oqgru.mongodb.net/?appName=Cluster0'
marc@Gumis-MacBook-Pro M-AC1API %
```

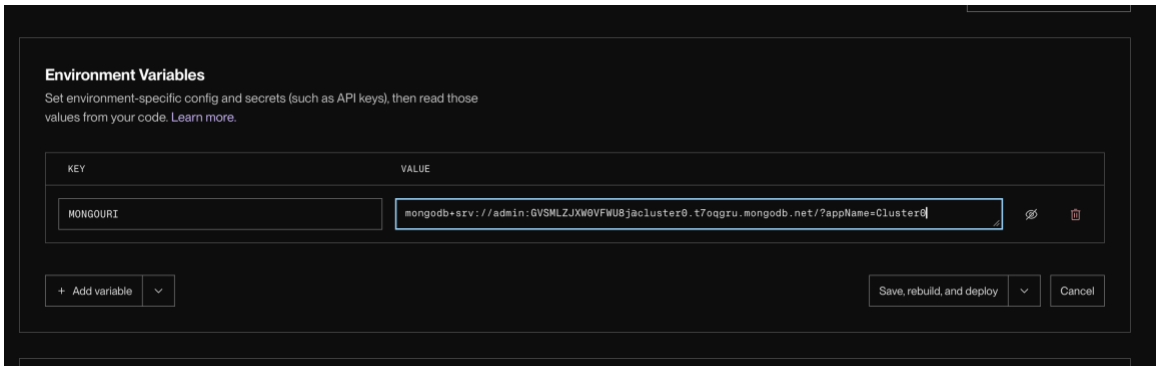
2. Se modifica config/config.js para leer la URI desde process.env.MONGOURI.

```
JS config.js M ×    JS products.js    JS app.js    main.scss
config > JS config.js > <unknown>
1   module.exports = {
2       mongoURI: process.env.MONGOURI
3   }
```

3. Se comprueba que la API sigue conectando correctamente a la base de datos.

```
To https://github.com/Rayo25Gumi/M-AC1API.git
   83f003c..593afae  main -> main
marc@Gumis-MacBook-Pro M-AC1API % export MONGOURI='mongodb+srv://admin:GVSMLZJXW0VFWU8j@cluster0.t7oqgru.mongodb.net/?appName=Cluster0'
marc@Gumis-MacBook-Pro M-AC1API % nodemon app.js
[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node app.js`
server*is*listening*on*port*3000
```

4. Se añade la variable de entorno MONGOURI al panel Environment de Render.



5. Se suben los cambios y se verifica que la API online sigue funcionando sin la URI expuesta al repositorio.

```
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node app.js`
server*is*listening*on*port*3000
^C%
marc@Gumis-MacBook-Pro M-AC1API % git add config/config.js
git commit -m "Move MongoDB URI to environment variable"
git push origin main
[main 5a92469] Move MongoDB URI to environment variable
 1 file changed, 2 insertions(+), 3 deletions(-)
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 381 bytes | 381.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Rayo25Gumi/M-AC1API.git
   593afae..5a92469  main -> main
marc@Gumis-MacBook-Pro M-AC1API %
```

Open file in editor (cmd + click)

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